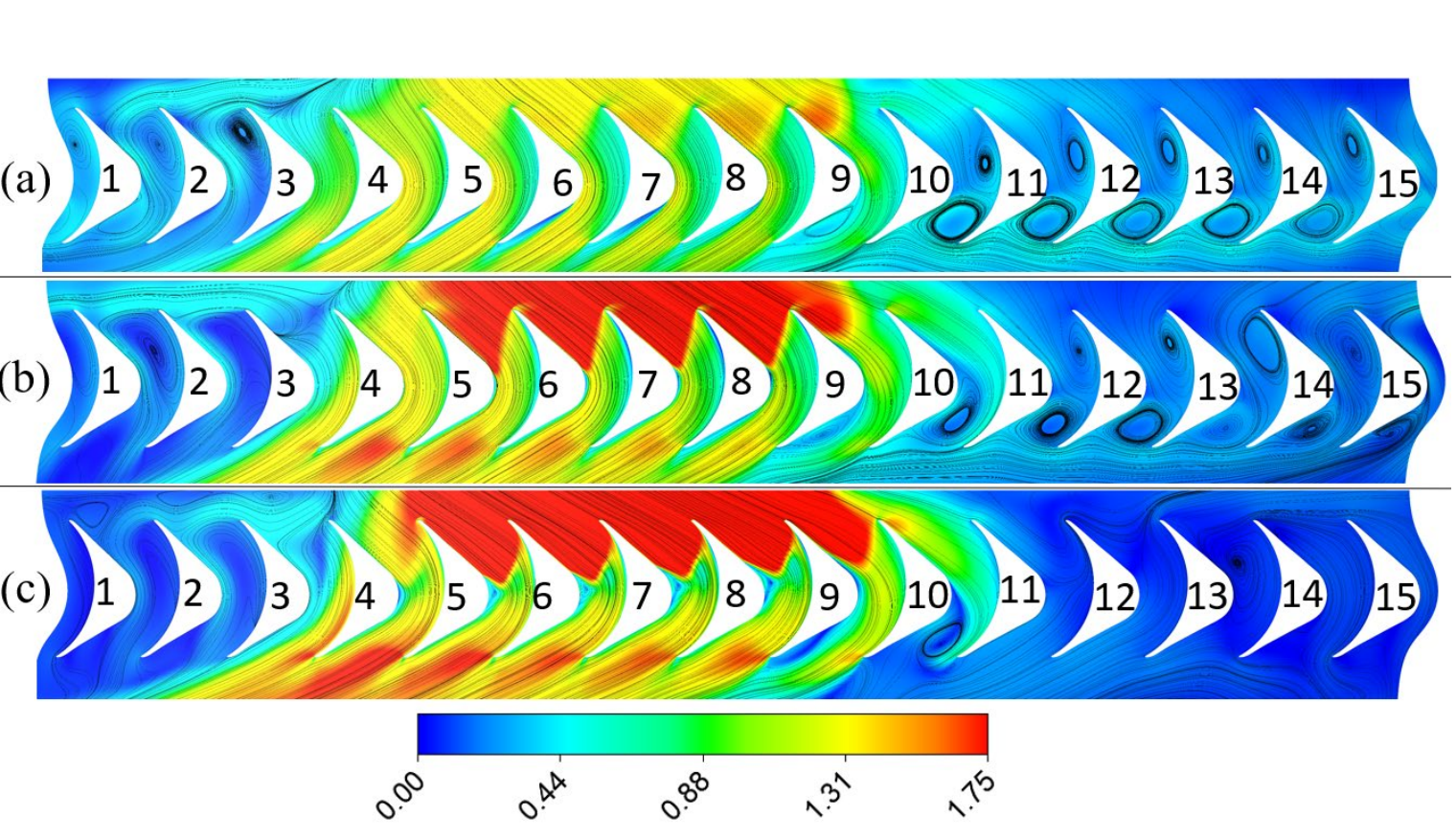
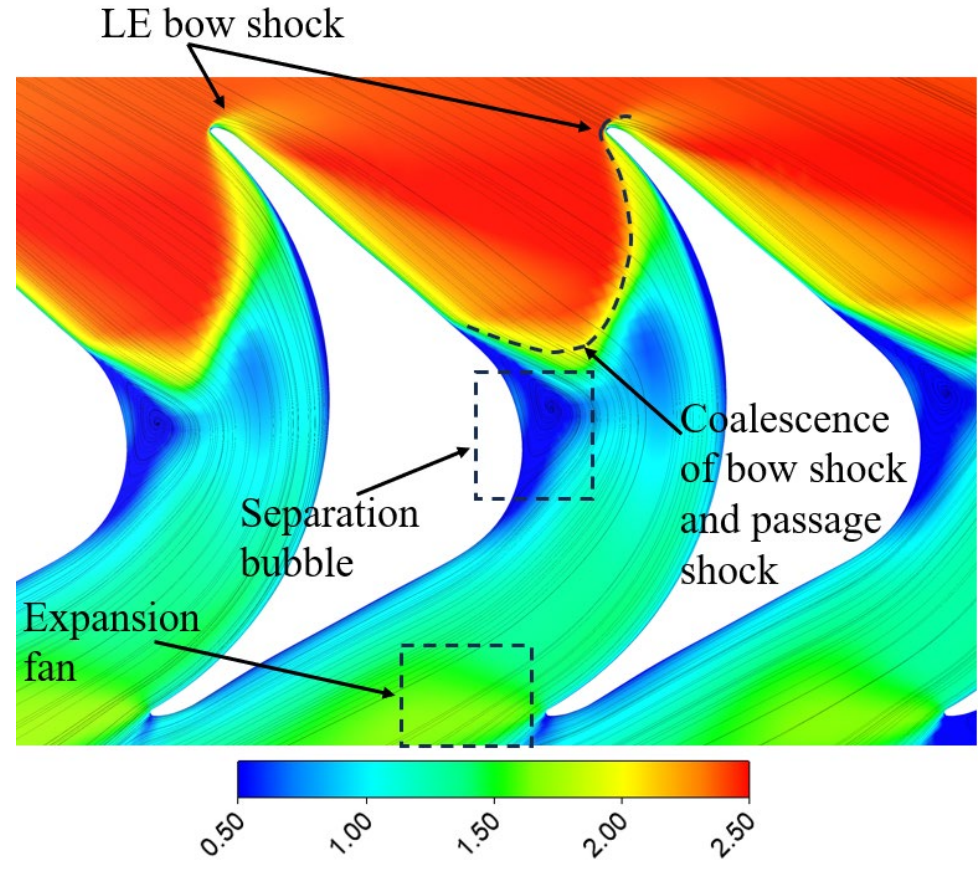

(a)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
(b)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
(c)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
0.00
0.44
0.88
1.31
1.75


LE bow shock
Coalescence
of bow shock
and passage
shock
Separation
bubble
Expansion
fan
0.50
1.00
1.50
2.00
2.50